Chih-Hsuan (Bella) Yang

Applied AI Research and Engineering

bellayang@anl.gov | chihhsuan-yang.github.io

GitHub | Hugging Face | Google Scholar | AgentsSci | LinkedIn

Postdoctoral researcher at Argonne building and evaluating large language model (LLM) agents. Expertise in reproducible harnesses, multi-agent systems, preference-based post-training, and agentic AI for Science.

Technical Skills

Python, PyTorch, Hugging Face, NumPy, pandas, pytest, Git. Agent evaluation, trace analysis, preference and reward modeling, reinforcement learning from human feedback (RLHF), and cost-aware inference.

Research and Industry Experience

Postdoctoral Researcher, Argonne National Laboratory — Jul 2025 - Present

- Build research harnesses for matched agent evaluation, trace analysis, and reproducible code and data.
- Mentor four students and coordinate collaborations from experimental design through papers and public releases.

Visiting Scientist, Corteva Agriscience — Sep 2024 - Jan 2025

- Trained preference models from scratch using experimental outcomes to ground rewards; incorporated scientist preferences when judgments were uncertain.
- Adapted multiple language models with Direct Preference Optimization (DPO) and Proximal Policy Optimization (PPO).

AI Resident, X, Google LLC — Sep - Dec 2023

- Developed stakeholder-based multi-agent LLM frameworks; studied information flow, coordination, communication, and bidding.

AI Graduate Researcher, Iowa State University — Jan 2020 - Jun 2025

- Co-led BioTrove and developed molecular language modeling and multi-fidelity methods for scientific discovery.

Selected Publications

Precise but Uncoupled — 2026
First author. NeurIPS Main Track, accepted.

Cost-Aware Protocol Routing — 2026
First author. Findings of EMNLP, published.

BioTrove — 2024
First author. NeurIPS Datasets and Benchmarks, Spotlight.

Education

Ph.D., Mechanical Engineering, Iowa State University — 2025
Co-major in Computer Engineering; minor in Applied Mathematics.

M.S., Computer Science, Iowa State University — 2025

B.S., Bioenvironmental Systems Engineering, National Taiwan University — 2019